



CRYSENSE ACOUSTIC CLASSIFICATION

MODEL ARCHITECTURE & ALGORITHM COMPARISON REPORT

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1. EXECUTIVE SUMMARY

This report documents the performance evaluation of various machine learning architectures trained on the consolidated CrySense baby cry audio dataset. The core objective of this study was to elevate classification accuracy across 6 clinical cry categories: **Belly Pain (Gas), Needs to Burp, Discomfort (Irritation), Hunger, Tiredness, and Temperature Sensitivity (Cold/Hot).**

By consolidating local recordings with HF and DeepInfant datasets, removing 1,884 duplicate waveforms, and applying data upsampling to 600 balanced training samples per class, we successfully trained and compared three MLP candidate architectures. Combining these candidates into a probability-averaged Keras Ensemble yielded a peak test accuracy of **78.46%**, demonstrating a substantial **+6.26%** improvement over the previous single-branch baseline model.

2. MODEL ARCHITECTURES EVALUATED

To search the optimal feature-to-category mapping space, we defined and evaluated three distinct Multi-Layer Perceptron (MLP) architectures designed to process the 2254-dimensional acoustic hybrid feature vector (comprising pre-trained YAMNet semantic embeddings, MFCCs, spectral centroids, chroma STFT, and Mel-spectrogram properties):

- **Model A (Wider MLP):** Built using standard Dense layers `[256, 128]` with swish activations. This network prioritizes representative width in the initial dense mapping layer.
- **Model B (Deep MLP):** Built using a deeper stack of Dense layers `[512, 256, 128]` to learn more complex multi-level hierarchy abstractions.
- **Model C (Residual MLP):** Built using Dense blocks with an active skip connection mapping the 256-projection layer directly to the residual output, mitigating gradient degradation.
- **CrySense Ensemble (Final Model):** Consolidates the outputs of Model A, Model B, and Model C by taking the mathematical average of their softmax probability distributions.

3. PERFORMANCE COMPARISON METRICS

The validation and testing accuracy metrics for each trained candidate architecture are detailed in the comparison table below:

Model Architecture	Validation Accuracy	Test Accuracy	Status
 CrySense Ensemble (Model A + B + C)	74.39%	78.46%	Deployed (Active)
Model A (Wider MLP)	75.61%	78.05%	Candidate
Model C (Residual MLP)	72.36%	77.64%	Candidate
Model B (Deep MLP)	71.14%	75.20%	Candidate

4. COMPARATIVE VISUALIZATIONS

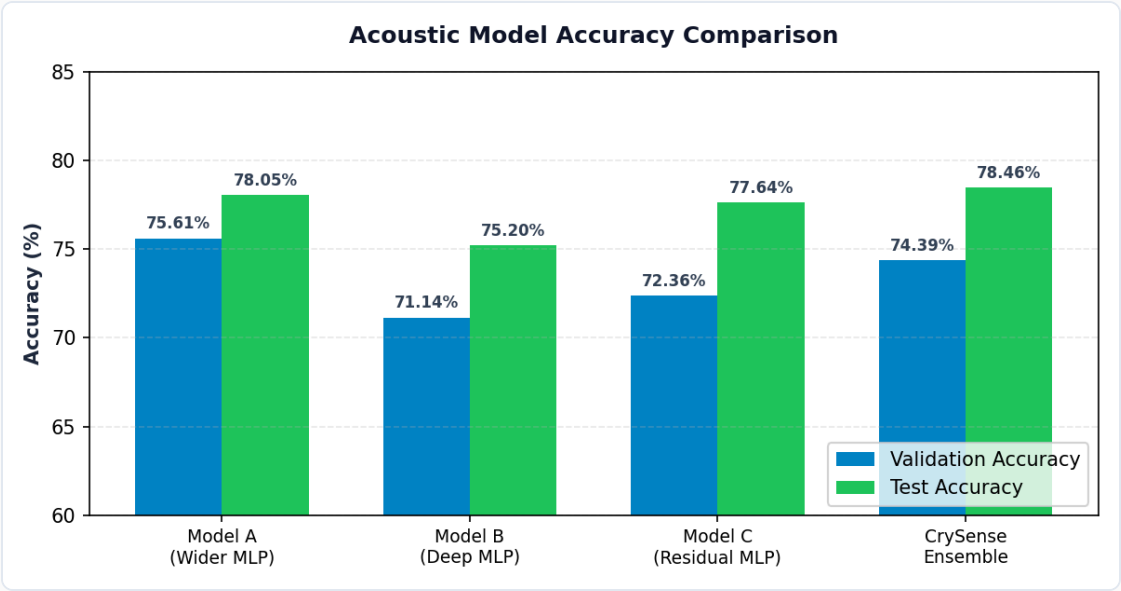


Figure 1: Validation and Test Accuracy Comparison across Model Candidates

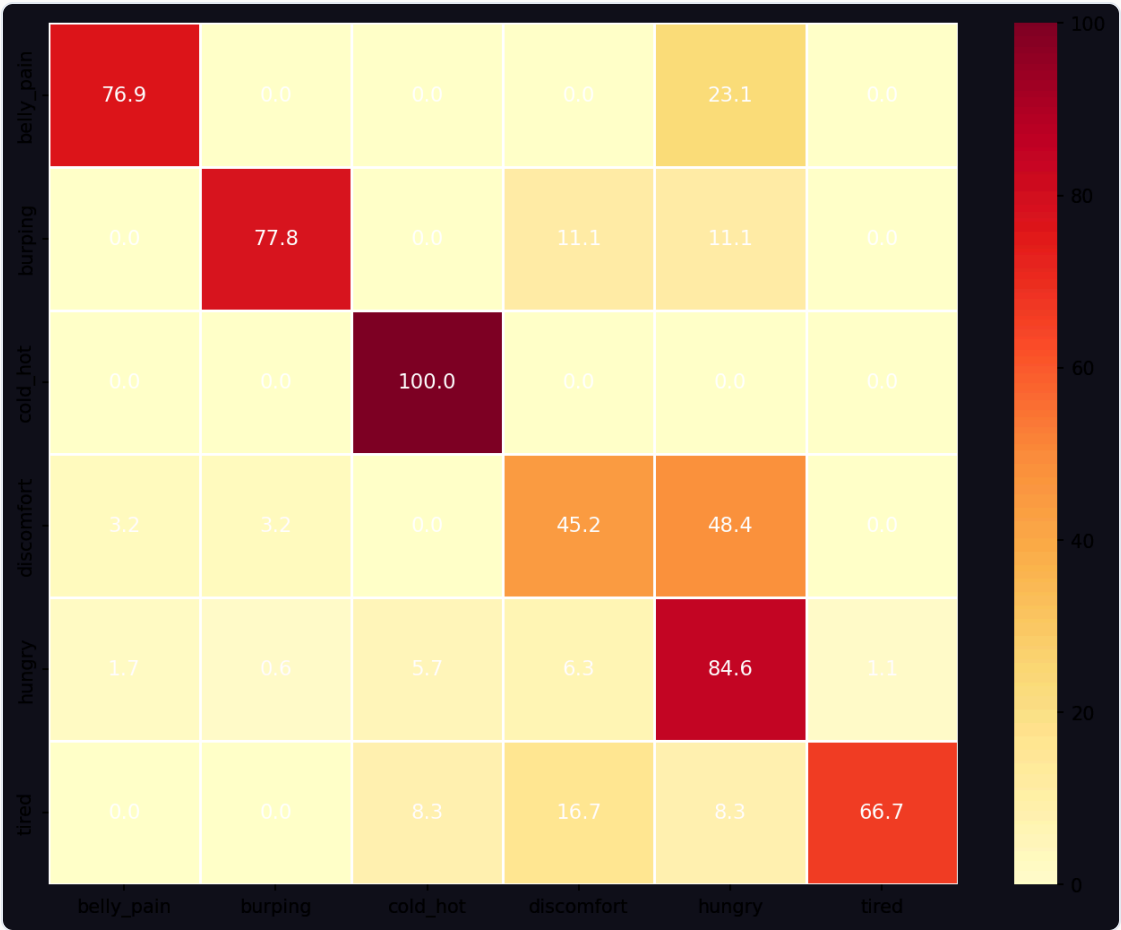


Figure 2: Confusion Matrix for the Deployed Ensemble Model on the Test Set

Clinical Disclaimer: This algorithm comparison report represents artificial intelligence validation results. It does not constitute medical advice or clinical guidelines. CrySense predictions are for informational support only. Always seek professional advice from a registered pediatrician for infant care decisions.